

Pathology: Pulmonary Vascular Disease



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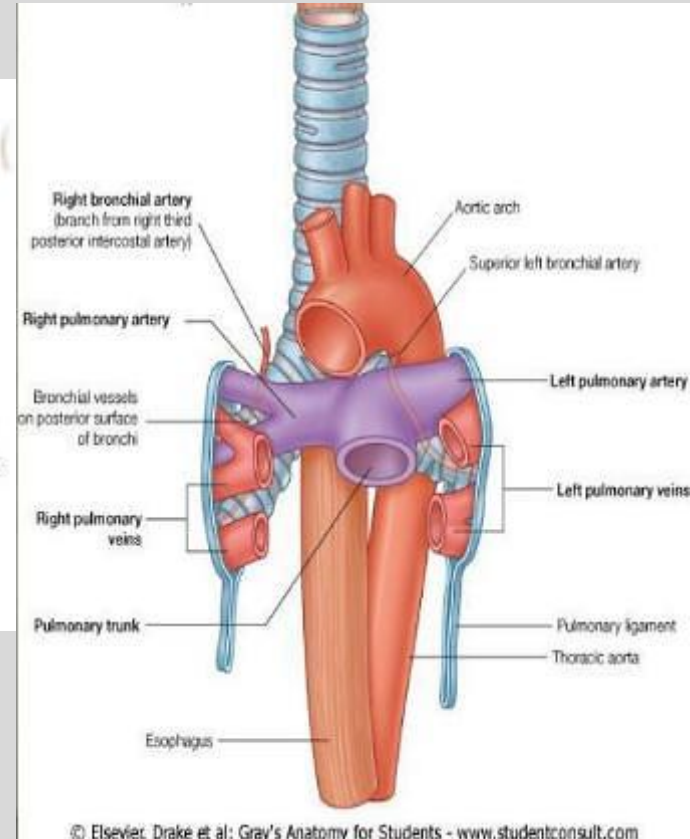
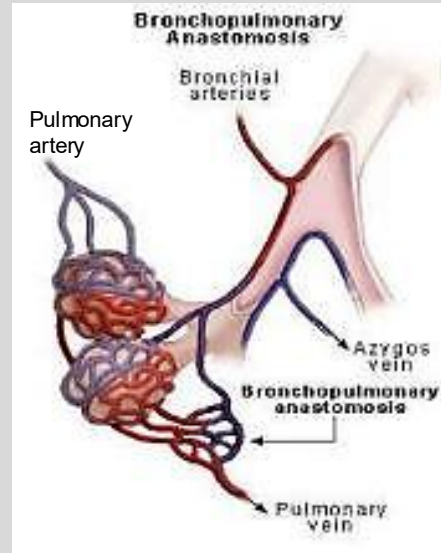
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Heal.
Innovate.
Reinvent the Future.

Objectives

1. Describe and compare the pathophysiology of **pulmonary embolism**, pulmonary **hemorrhage** and **infarct**
2. Discuss the etiology, pathophysiology, lesions, and grading of **pulmonary hypertension**.
3. Interpret the pathophysiology and associated pulmonary lesions of **Goodpasture syndrome**.
4. Specify how the pathophysiology and associated pulmonary lesions of **granulomatosis with polyangiitis** (aka Wegener granulomatosis) differ from similar diseases.

Lung circulation review

- **Pulmonary** arteries: From **right ventricle**, perfuse alveoli, **deoxygenated** blood
- **Bronchial** arteries: From the **left ventricle** as aorta & intercostal artery branches, **perfuse** conducting airways
 - Bronchial capillary blood drains to **bronchopulmonary anastomoses** that empty to pulmonary, azygos, and intercostal veins to right atrium



Pulmonary embolisms (PE)

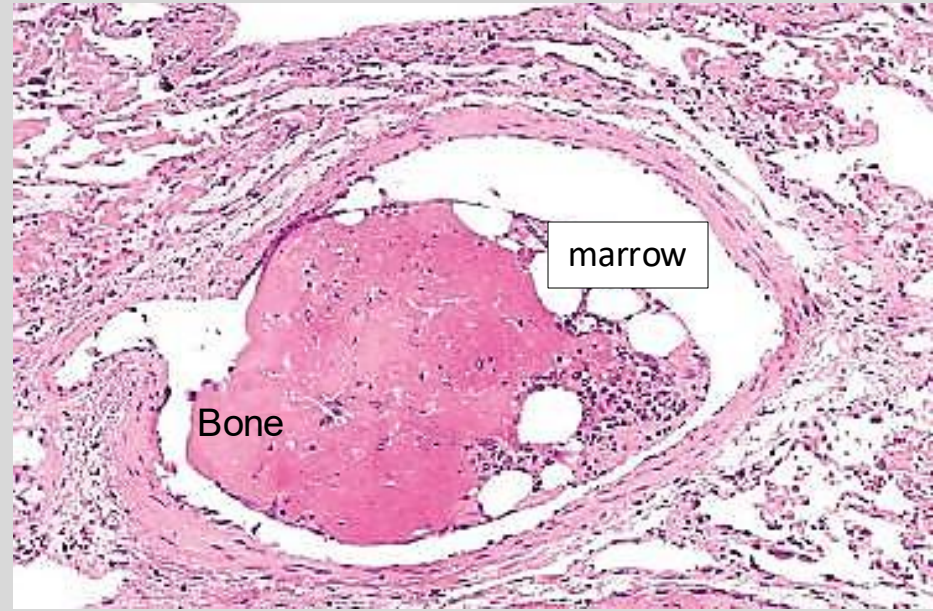
- 3rd most common acute cardiovascular condition worldwide
 - Rare in pediatrics, common after age **75 years**
 - Also, risk for **inpatients** > general population
- **Most** clots occluding large arteries → **embolic**, usually **deep leg vein thrombi** (95%), usually **femoral vein**
 - Rarely have infected (septic infarct), **fat**, air, amniotic fluid or **neoplastic** embolus
 - Example: Small bone **marrow** emboli after chest compressions during **resuscitation** (often in those who die)
- **Usually** in patients with **predisposing conditions** increasing the risk of thrombosis:
 - **Primary** hypercoagulable states: Factor V Leiden, antiphospholipid syndrome, prothrombin **mutations**
 - **Secondary** hypercoagulable states: Cardiac disease, obesity, **cancer**, **pregnancy**, oral **contraceptives**, **post-op**, **immobilization**/bed rest, patients with hip **fractures** (particularly high risk), indwelling central venous lines (nidus for right atrial thrombi formation, can embolize to lungs)
- **Virchow's Triad** (general factors contributing to thrombosis): **Hemodynamic changes** (stasis, turbulence), **endothelial injury**, **hypercoagulability**

PE: Pathophysiology

Two main pathophysiologic **consequences**:

- **Respiratory compromise** due to the **nonperfused**, although ventilated, segment
- **Hemodynamic compromise** due to increased resistance to pulmonary blood flow caused by embolic **obstruction** → may develop **sudden death**, acute right-sided heart failure (acute **cor pulmonale**), or pulmonary **hypertension**

Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 15 & fig .16

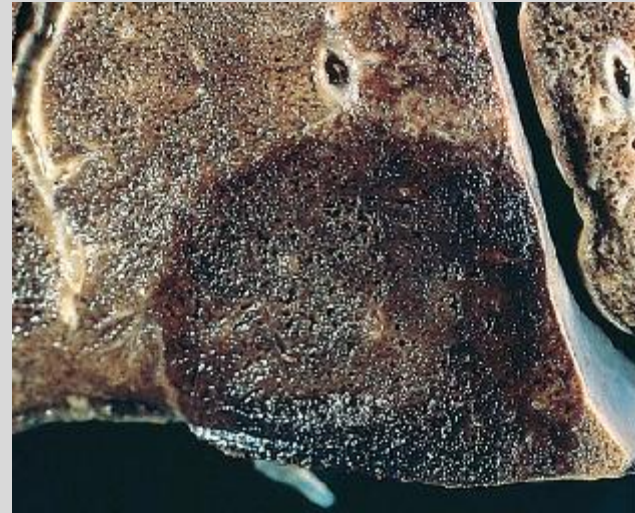


PE: Hemorrhage and infarction

- **Large emboli** lodge in main pulmonary artery, major branches or the bifurcation (**saddle embolus** - may cause pulmonary infarction (tissue death), but **immediate danger** is hemodynamic collapse (shock and cardiovascular failure))
- **Smaller emboli** travel out into **peripheral** vessels
 - May cause **hemorrhage or infarction**
- With **compromised cardiovascular** function, such as heart or lung disease, **infarction** more likely
 - Pulmonary infarct is **classically hemorrhagic**
 - Histologically, hemorrhagic area shows **ischemic necrosis** of alveoli, bronchioles, and vessels



Saddle embolus from the femoral vein



Acute hemorrhagic pulmonary infarction

PE: Embolism → infarction

- ~ 10% of emboli cause infarction
 - ~ 75% **lower lobes** (where perfusion > ventilation)
 - > half, **multiple** lesions - from barely visible to large parts of a lobe
- PE distinguished from clots that form after death (postmortem) by **lines of Zahn** →
 - Grossly, postmortem clots have dark-red “**currant jelly**” part, and a yellow “**chicken fat**” part, and usually are not attached to underlying vessel wall
 - Postmortem clots are sometimes mistaken for antemortem venous thrombi
 - Antemortem venous thrombi can have lines of Zahn, but typically more prominent and frequently observed in sites of rapid blood flow like the heart and arteries
- Red cells lyse in 48 hours → infarct becomes paler → then red-brown (hemosiderin) → eventually, contracted **scar**



H&E stained thrombus: Recent pulmonary thromboembolus exhibiting **prominent lines of Zahn**
https://upload.wikimedia.org/wikipedia/commons/thumb/1/16/Pulmonary_thromboembolus%2C_recent_%283626658803%29.jpg/500px-Pulmonary_thromboembolus%2C_recent_%283626658803%29.jpg

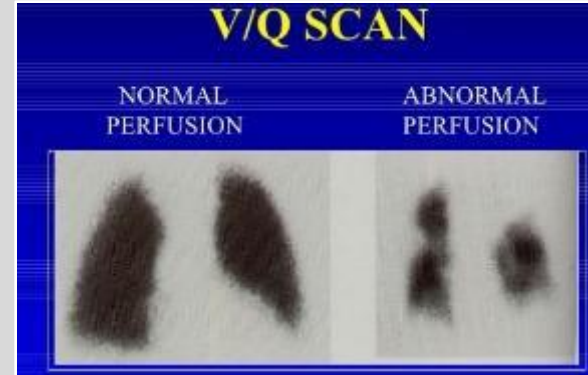
PE: Clinical findings

- 60-80% of PE clinically **silent**
 - **Small** emboli may be **silent** or only have **transient** chest pain and cough
 - ~5% cause **acute cor pulmonale**, shock, or **death** (typically large “saddle emboli”)
 - During CPR, patients frequently have electromechanical dissociation (**EMD**- EKG has rhythm, but no pulses palpated because no blood is entering the pulmonary artery circulation)
 - If patient **survives**, condition may clinically **mimic MI** (severe chest pain, dyspnea, and shock)
 - In remaining patients with symptomatic PE, cause symptoms of **ventilation-perfusion (V/Q) mismatch** (when airflow and blood flow in the lungs are not balanced), and/or pulmonary **infarction** - particularly **dyspnea** and **pleuritic** chest pain
 - Most commonly find: Dyspnea, pleuritic pain, **pleural friction rub** (**overlying fibrinous pleuritis**) and **cough**; may have **fever**, **hemoptysis** (with emboli producing infarction), and calf or thigh swelling or pain (half of cases)*
- Recurrent small emboli can give rise to **pulmonary hypertension**

*A positive Homans's sign (calf pain with dorsiflexion of the foot) may be associated with the presence of thrombosis

PE: Lab

- Useful **screening** test → **normal D-dimer level excludes PE**
 - In **hemodynamically stable** patients (low-moderate pretest probability)
- **Definitive** diagnosis usually with **CT pulmonary angiogram (CTPA)** or **MRPA** (identifies obstructed pulmonary arteries)
- Rarely, other diagnostics like **ventilation-perfusion (V/Q)** scanning required
- Deep vein thrombosis (DVT for example of the legs) can be diagnosed with **duplex ultrasonography/ultrasound**
- Chest radiography/chest x-ray (**CXR**) is **neither sensitive nor specific** for PE & may be normal or disclose a **pulmonary infarct**



Patient inhales radioactive gas (**V**) and injected with radioactive tracer (**Q**), gamma camera (scintillation camera) detects gamma radiation

<https://th.bing.com/th/id/R.16a3f02c4ead6de94607561cb097b264?rik=9qOIPKzG3dyTFw&riu=http%3a%2f%2fimage.sldesharecdn.com%2fgerberpulmonaryembolism-1233842848283730-3%2f95%2fgerber-pulmonary-embolism-23-728.jpg%3fcb%3d1233821346&ehk=FA8M9NS44EopusBuzObhCL4ChrsmPIFVM574xgeiHI%3d&risl=&pid=ImgRaw&r=0>

PE: More lab → D-dimers

- D-dimers are a **fibrin** degradation product
 - From blood clot degradation (**thrombolysis** - usually refers to medical treatment) by **fibrinolysis** (the biologic process)
 - Blood levels help diagnose **thrombosis**
- Indicate the **presence** of a clot but does **not** specify its **location or type** (whether forming an artery or a vein)
- **Negative** D-dimer does not change probability of PE in people at **high risk** for thrombosis
- D-dimers usually **high** in people **with PE**, but **not specific**:
 - May also occur in **DVT**, **malignancy**, **sepsis**, acute **coronary syndrome**, acute aortic syndrome, **trauma**, recent **surgery** and **pregnancy** for example

PE: Prevention/treatments

- After initial acute insult, esp. **young** patients, emboli often **resolve** via contraction & fibrinolysis
- If **unresolved**, multiple small emboli may lead to **pulmonary hypertension & chronic cor pulmonale**
- High risk of **recurrence** → recurrent embolism may → **pulmonary hypertension & cor pulmonale**
- Important to note is that a **small PE may predict a larger one**
- **Prophylactic** therapy (to prevent blood clots like DVT)
 - **Early ambulation** postoperative & postpartum patients
 - Compression **stockings**/hose
 - **Anticoagulation in high-risk** individuals
- Treatment includes **anticoagulation & supportive** measures
 - If risk of recurrence & anticoagulation contraindicated: **Inferior vena cava filter**

Pulmonary hypertension definition: Mean **pulmonary artery pressure ≥ 25 mm Hg at rest**

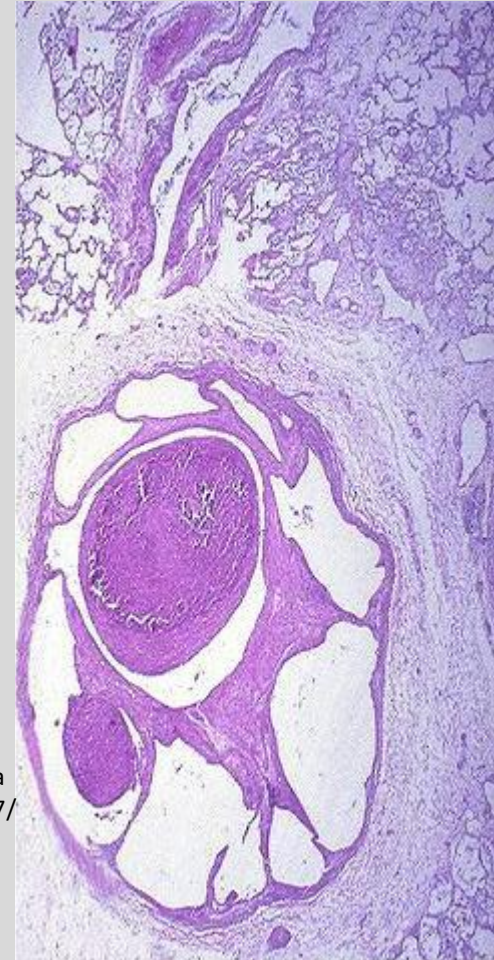
- Direct disease of lung arteries
- PH is a group category based on underlying mechanisms, **WHO classified** PH into five groups that have diverse **causes** even within each group:
 1. Primary pulmonary hypertension (aka **idiopathic** pulmonary arterial hypertension, PAH) – diverse collection, all mostly effect **small** pulmonary muscular **arteries**
 2. Due to **left heart failure** - (most frequent cause of overall PH)
 3. Due to **lung diseases** and/or **hypoxia**
 4. Chronic **thromboembolic** pulmonary hypertension and other **obstructions**
 5. Unclear and/or **multifactorial** mechanisms

Pulmonary hypertension **common causes are secondary causes:**

- Most cases associated with **structural** cardiopulmonary conditions that
 - ↑ pulmonary blood **flow**
 - ↑ pulmonary vascular **resistance**,
 - **left heart** blood flow **resistance**
- **Chronic obstructive or interstitial lung diseases** ↑ pulmonary resistance
 - Ex. **Recurrent thrombo-emboli**
 - Ex. **Autoimmune** diseases - notably **systemic sclerosis** involving pulmonary vessels
- Antecedent congenital or acquired **heart disease**
- **Obstructive sleep apnea** - common, associated with obesity and hypoxemia

- Fibrous bands from a healing thromboembolus

- <https://www.memoranga.com/flashcards/47837/Respiratory+Pathology/>

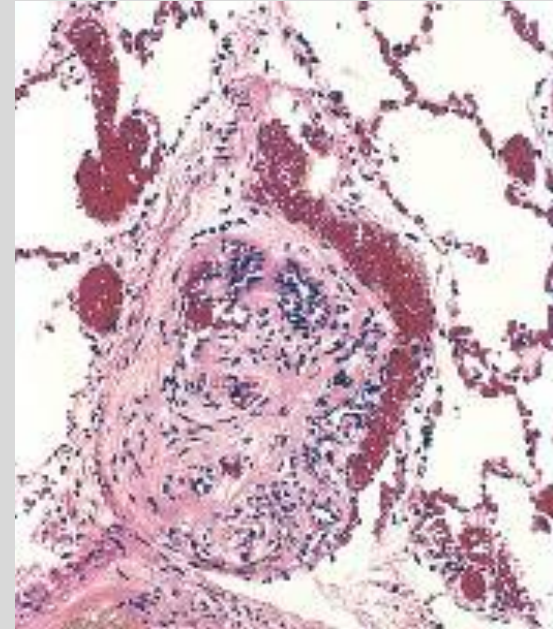


Primary pulmonary hypertension - pathogenesis

- Much learned by investigating molecular basis of **uncommon familial forms**:
 - **First mutation** was in germline gene mutation for bone morphogenetic protein receptor type 2 (**BMPR2**)
 - **Inactivating** → insufficiency of BMPR2 → dysfunction & **proliferation** of **endothelial & vascular smooth muscle** cells)
 - 75% of familial PH and 25% of sporadic cases have this
 - **Other mutations** have been discovered that also **converge on BMPR2 pathway** and affect intracellular signaling

Pulmonary hypertension: Histopathology

- All forms are associated with **medial** (tunica media) **hypertrophy** of pulmonary muscular and elastic arteries & right ventricular hypertrophy
 - **Arterioles & small arteries** affected by medial hypertrophy & intimal **fibrosis**
- An extreme in pathologic changes is the **plexiform lesion**



H&E: **Plexiform** lesion of small arteries that is characteristic of advanced pulmonary hypertension

Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 15, fig 15.25

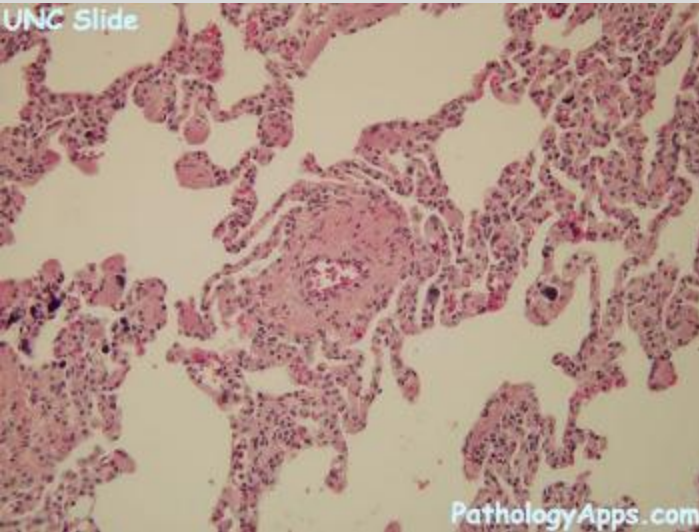
Grade*	Histologic Changes in Pulmonary Arteries and Arterioles
1	Medial hypertrophy without intimal changes
2	Medial hypertrophy with <u>intimal cellular proliferation</u>
3	Medial hypertrophy, <u>intimal proliferation</u> , and <u>intimal fibrosis</u>
4	Progressive generalized vascular dilatation and occlusion by intimal fibrosis and fibroelastosis; appearance of <u>plexiform lesions</u>
5	Appearance of veinlike branches of hypertrophied muscular arteries, cavernous lesions, and <u>angiomatoid lesions</u>
6	<u>Necrotizing arteritis</u>

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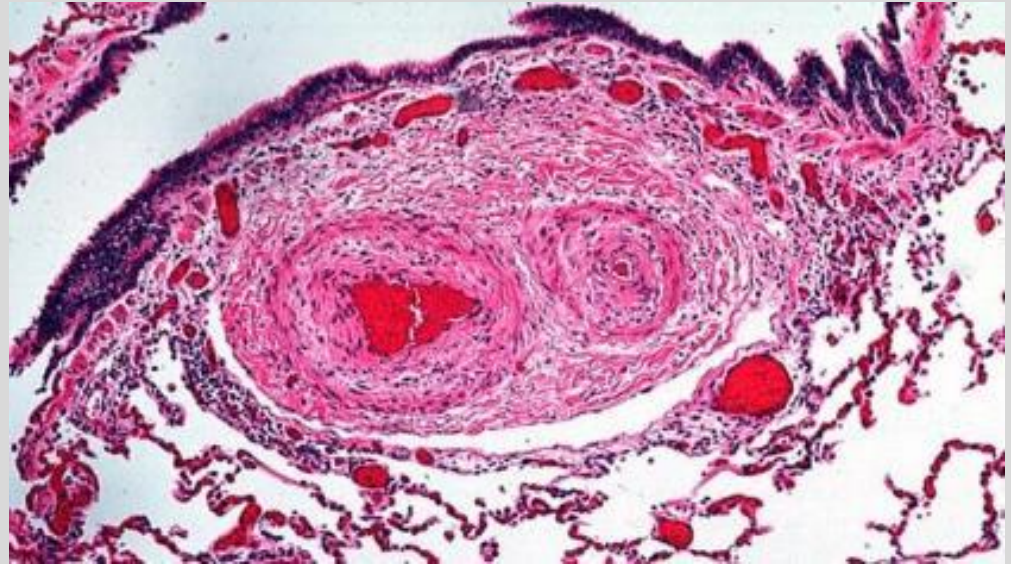
- *Grades 1 and 2 - mild, potentially reversible disease
- Grade 3 - disease with intermediate severity and borderline reversibility
- Grades 4 through 6 - irreversible disease

<https://pubs.rsna.org/doi/full/10.1148/rg.307105710>

Pulmonary hypertension: Grades



- Grade 1 PH
- Early muscular medial **hypertrophy**
- But no intimal proliferation or fibrosis



- Grade 3 PH
- **Intimal fibrosis**
- Narrowing vessel lumens
- https://peir.path.uab.edu/library/_data/i/upload/2013/08/01/20130801100611-ec90d18f-me.jpg

Pulmonary hypertension: Diagnosis

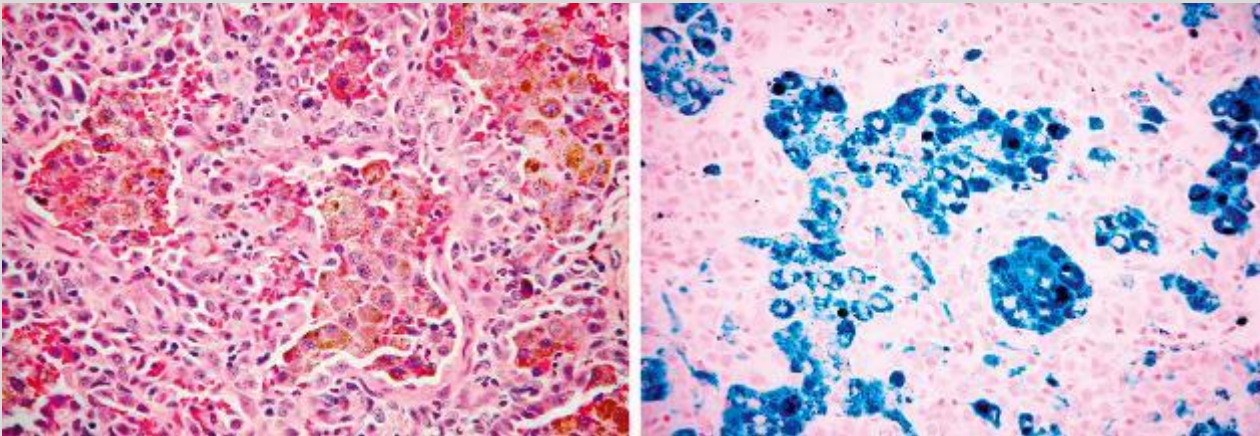
- **ECG:** Includes RVH, with right-axis deviation
- **CXR:** RV & main pulmonary arteries enlargement with rapid tapering of distal vessels
- **Echocardiogram:** Dilated pulmonary artery, RA & RV hypertrophy & dilation
- Right atrial **catheterization**, gold standard that directly measures pressure in lung arteries, shows **increased pulmonary artery pressure**

Pulmonary hypertension: Clinical findings & treatments

- Signs & symptoms in all types typically evident **only in advanced disease**
 - Early, mild, subtle signs: Shortness of breath during exercise or slight tiredness
- In **idiopathic** disease, usually presents w/ **dyspnea & fatigue**, some have **anginal** chest pain → severe **respiratory distress**, **cyanosis**, syncope, peripheral **edema**, jugular vein distention (**JVD**), right ventricular hypertrophy (**parasternal lift**) with severe disease → death from decompensated **cor pulmonale**, often with superimposed thromboembolism & pneumonia, ensues within 2-5 years in ~80% of patients
- In general PH is most common in **older adults** (often > 65 y.o., with ↑age & underlying heart or lung disease) & women slightly > men
 - Sub-types like **PAH** frequently (60-80%) in **younger** 30-60 y.o. & middle-aged women, rarely children
- Treatment depends on **underlying cause**
 - Vasodilators used in PAH
 - Secondary disease: Therapy is directed at trigger (ex., thromboembolic disease or hypoxemia)
 - Lung transplantation: Definitive treatment (select patients)

Diffuse pulmonary hemorrhage syndromes include

- **Goodpasture syndrome**
- Vasculitis-associated (includes hypersensitivity angiitis, **polyangiitis with granulomatosis**, lupus)
- **Idiopathic pulmonary hemosiderosis**
- Others (ex. CHF)



Diffuse **pulmonary hemorrhage**: Intra-alveolar spaces with large numbers of **macrophages laden** with phagocytosed red cells and hemoglobin breakdown product **hemosiderin (siderophages)** - **Prussian-blue** stain identifies **iron** derived from hemoglobin in macrophages

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Goodpasture syndrome involvement of the lung is an anti–glomerular basement membrane antibody disease with pulmonary involvement:

- Uncommon **autoimmune** disease: **Kidney & lung injury** is caused by **circulating autoantibodies** against **noncollagenous** domain of the **$\alpha 3$ chain** of **collagen IV**
- Most cases in **teens or 20s**, **male** preponderance (unlike most autoimmune diseases) & majority active **smokers**
- Called anti–glomerular basement membrane disease if only renal disease

Goodpasture syndrome

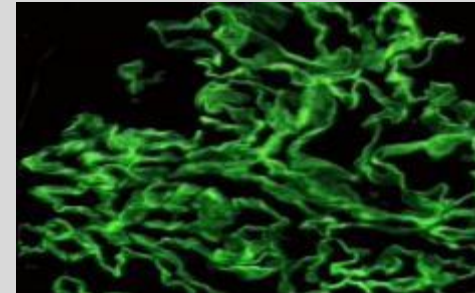
- Pathogenic antibodies cause **inflammatory destruction** of **basement membranes** in **lung** alveoli and kidney glomeruli
 - Presents with necrotizing hemorrhagic **interstitial pneumonitis** & usually **rapidly progressive glomerulonephritis**
- **Trigger** that initiates production of anti–basement membrane antibodies is **unknown**
- As with other autoimmune disorders, there is an association with **HLA subtypes** (e.g., HLA-DRB1*1501 & DRB1*1502)

Goodpasture syndrome: Clinical findings

- At autopsy, **lungs heavy**, with areas of red-brown **consolidation**
- Histopathology: Focal necrosis of alveolar walls associated with **intra-alveolar hemorrhages**; often alveoli contain **hemosiderin-laden macrophages**
- **Later** stages have possible **fibrous thickening** of septa, **hyperplasia** of type II pneumocytes
- Kidneys show focal **proliferative glomerulonephritis** in early cases, or **crescentic glomerulonephritis** in patients with **rapidly progressive glomerulonephritis**



- Diffuse pulmonary hemorrhage in Goodpasture syndrome ([Case courtesy of Yale Rosen, MD
- https://www.flickr.com/photos/pulmonary_pathology/3731499661



Immunofluorescent microscopy - **linear deposits** of immunoglobulins along basement membranes of the septal walls

Goodpasture syndrome: Clinical findings & treatments

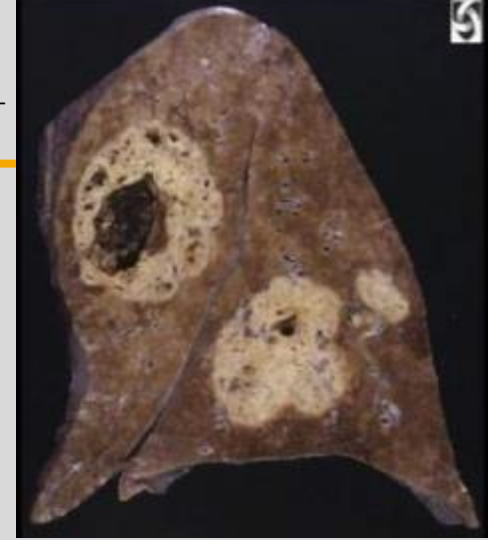
- **Usually begins** with respiratory symptoms (**hemoptysis**), CXR shows evidence of **consolidations**
- Soon **after respiratory** symptoms: **Glomerulonephritis** manifestations occur and lead to **rapid progressive renal failure**
- Most common cause of death is **uremia**
- Rx: **Corticosteroids**, Plasmapheresis & **Cyclophosphamide**
- Intensive **plasmapheresis** removes anti–basement membrane antibodies and other possible mediators of immunologic injury
- Simultaneous **immunosuppressive therapy** inhibits further antibody production, improving both lung hemorrhage and glomerulonephritis

Granulomatosis with polyangiitis (GPA)

- **Autoimmune** disease that **mostly** involves **upper respiratory tract** and/or **lungs**
 - **c-ANCA 95% (PR3-ANCA)**
- **Hemoptysis** common presenting symptom
- Childhood to middle age, peak incidence ~ 40 y.o., M>F
- A transbronchial lung biopsy might provide the only tissue available for **diagnosis**
 - Important histologic features are **capillaritis** and scattered, poorly formed **granulomas**
 - Characteristically **necrotizing vasculitis** with the following:
 - Acute **necrotizing granulomas** of **upper respiratory tract** and/or **lungs**
 - Granulomas have a **geographic** pattern of **necrosis**
 - Necrotizing or granulomatous **vasculitis** of **small-medium** sized **vessels**, mostly in lungs
- If **renal disease** is present: Proliferative or necrotizing, focal, crescentic **glomerulonephritis**

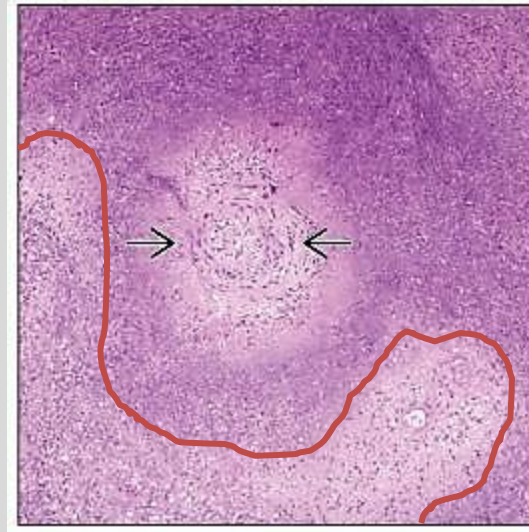
GPA pathology

<https://image.slidesharecdn.com/mb4-yearreview-term1-131110212733-phpapp01/95/pathology-reviewterm1-42-638.jpg?cb=1384127067>



- Lung grossly showing **cavitating** grey white **lesions**
- Histologically: Irregularly shaped **geographic necrosis** w/ epithelioid cells & giant cells forming **granulomas** &
 - **Leukocytoclastic vasculitis** (neutrophils)

Classic **histologic triad** includes **vasculitis**, **granulomas**, tissue **necrosis**



H&E stain: Geographic necrosis with a basophilic smudgy appearance surrounds an ablated vascular space

GPA: Pathogenesis & treatment

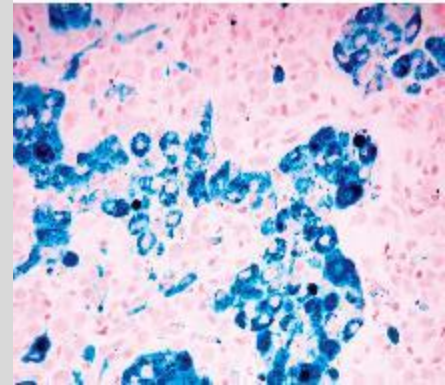
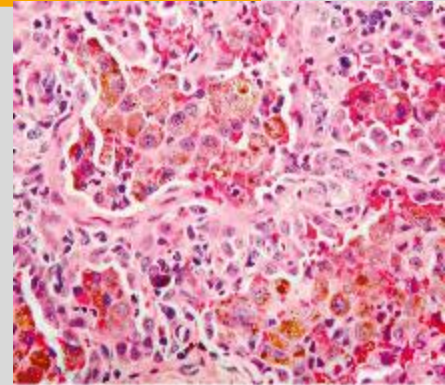
- Most likely T cell–mediated **hypersensitivity** response to normally “**innocuous**” **inhaled microbial** or **other environmental agents**
 - Supported by presence of **granulomas** & dramatic **response to immunosuppressive therapy**
- **Necrotizing medium-sized & small vessel vasculitis** involving:
 - **Upper respiratory tract**
 - **Lungs**
 - **Renal vessels**
- Treatment is similar to Goodpasture syndrome

Idiopathic pulmonary hemosiderosis

- **Rare** disorder with **intermittent, diffuse** alveolar hemorrhage, mostly in young **children**
 - M>F, peak incidence ~27 years
- **Cause & pathogenesis are unknown** - But, **favorable response** to long-term **immunosuppression** with prednisone and/or azathioprine indicates that an **immunologic** mechanism may be involved
- Defined by **recurrent alveolar bleeding w/o** an identifiable **cause**
- Usually **presents** with an **insidious** onset of productive **cough**, **hemoptysis**, and **anemia** associated with **diffuse bilateral pulmonary infiltrates**
- **Diagnosis of exclusion** - rule out all other causes first

Idiopathic pulmonary hemosiderosis - Histopathology

- The lungs are **increased in weight** due to areas of red to red-brown **consolidation**
- Alveolar spaces contain numerous **siderophages**, alveolar septa are thickened
- **Histopathological hallmark: Hemorrhage** into alveolar spaces and **hemosiderosis**, **siderophages** within alveoli; **necrosis** of alveolar walls
- Hyperplasia of type II pneumocytes and some degree of interstitial fibrosis may be present
- **No** vasculitis, capillaritis, or **inflammation**



Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 15, fig 15.26

Summary

- Pulmonary **embolism/infarct** – pulmonary infarcts are usually **hemorrhagic** because of dual supply of lungs
- Pulmonary **hypertension** – can be genetic in some cases (**Primary idiopathic**) associated with BMPR2 germline mutation; for other causes see **WHO** classification
- **Goodpasture** syndrome – affects **lungs & kidneys**; **autoimmune**, usually **male**; circulating **auto-antibodies** against **collagen IV**
- **Granulomatosis with polyangiitis** – autoimmune, **95% PR3-ANCA**; **necrotizing vasculitis with triad** of necrosis, granulomas & vasculitis of upper respiratory tract and/or lungs small to medium-sized vessels (mostly in lungs), and **renal** disease (**glomerulonephritis**)
- **Idiopathic pulmonary hemosiderosis** – rare, **children**; **intermittent** diffuse alveolar hemorrhage & hemosiderosis, siderophages, necrosis of alveolar walls, responds to **immunosuppressants**, **Dx of exclusion**

Thank you,
Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>